BioSysteM Research Retreat in Ettal

My road to founding and agentic bioengineering

by Tim Kucera

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Founder trajectory

Wet lab, engineering, and modeling in search for programmable biology



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EKU Tübingen
Biology

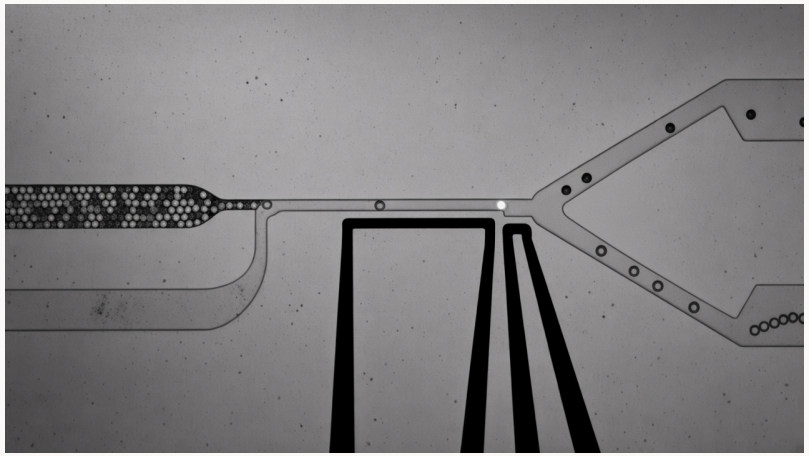
MSc
ETH Zürich
Biotechnology

2021
DropSort
Droplet Sorting Software

PhD
ETH / MPI
Molecular Machine Learning

2025
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AI infrastructure for Biotech

2026
Accubator
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High-throughput droplet sorting on a microfluidic chip

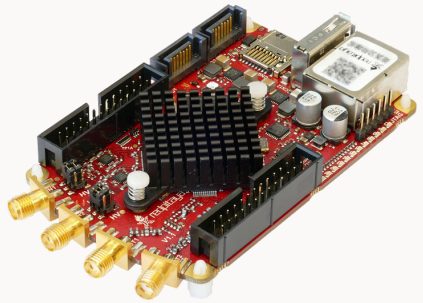
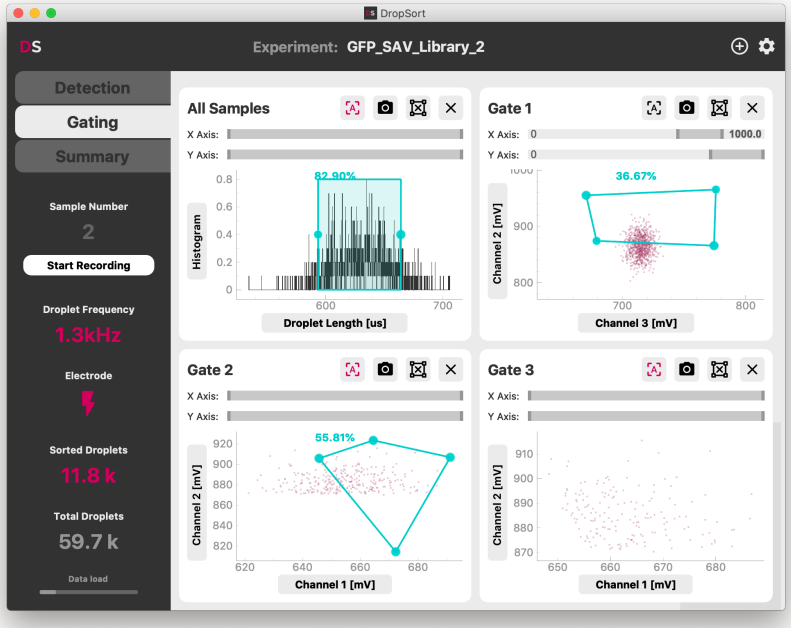


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A high-performance data processing card (FPGA)



DropSort software for running FPGA-based fluorescence activated droplet sorting.

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Image adapted from Goodfellow et al. (2014)

Generative modeling for protein design and engineering

ProteinShake

Kucera et al. (2023)

bioverse

Kucera et al. (unpublished)

Frameworks for standardizing biological data for machine learning analyses

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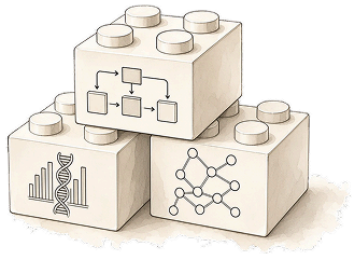
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Public databases,
papers, models

The Imaginary Platform



Standardized biological **data**, machine learning **models**,
and processing **workflows**



ML Engineers
Tools for developing models
Accelerating development cycles



Biotech Researchers
Graphical user interfaces
Accessible models for non-technical users

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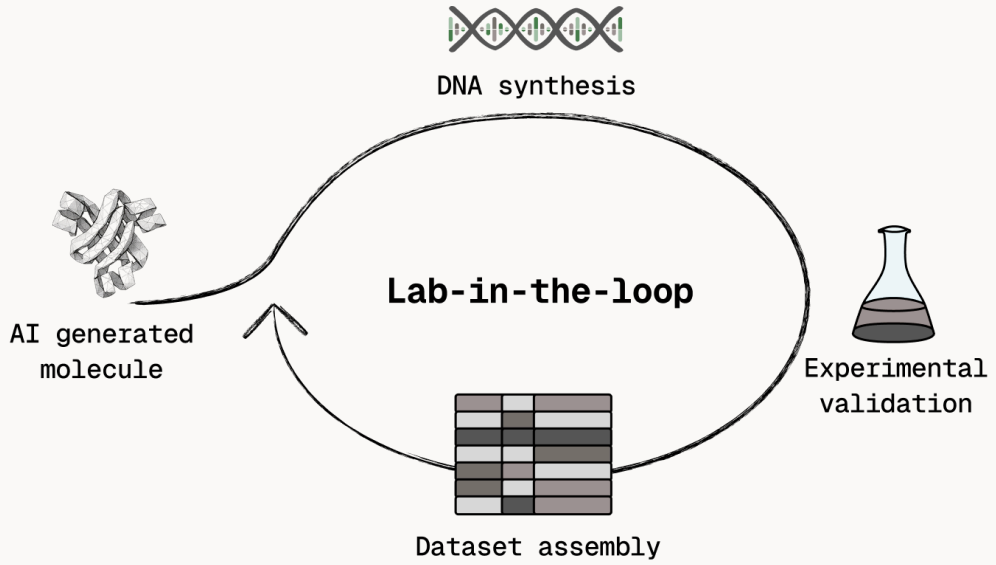
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More information at: www.bio-m.org/accubator

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Thinking about a company? Just do it!

Much of what you are doing as a researcher has a market value

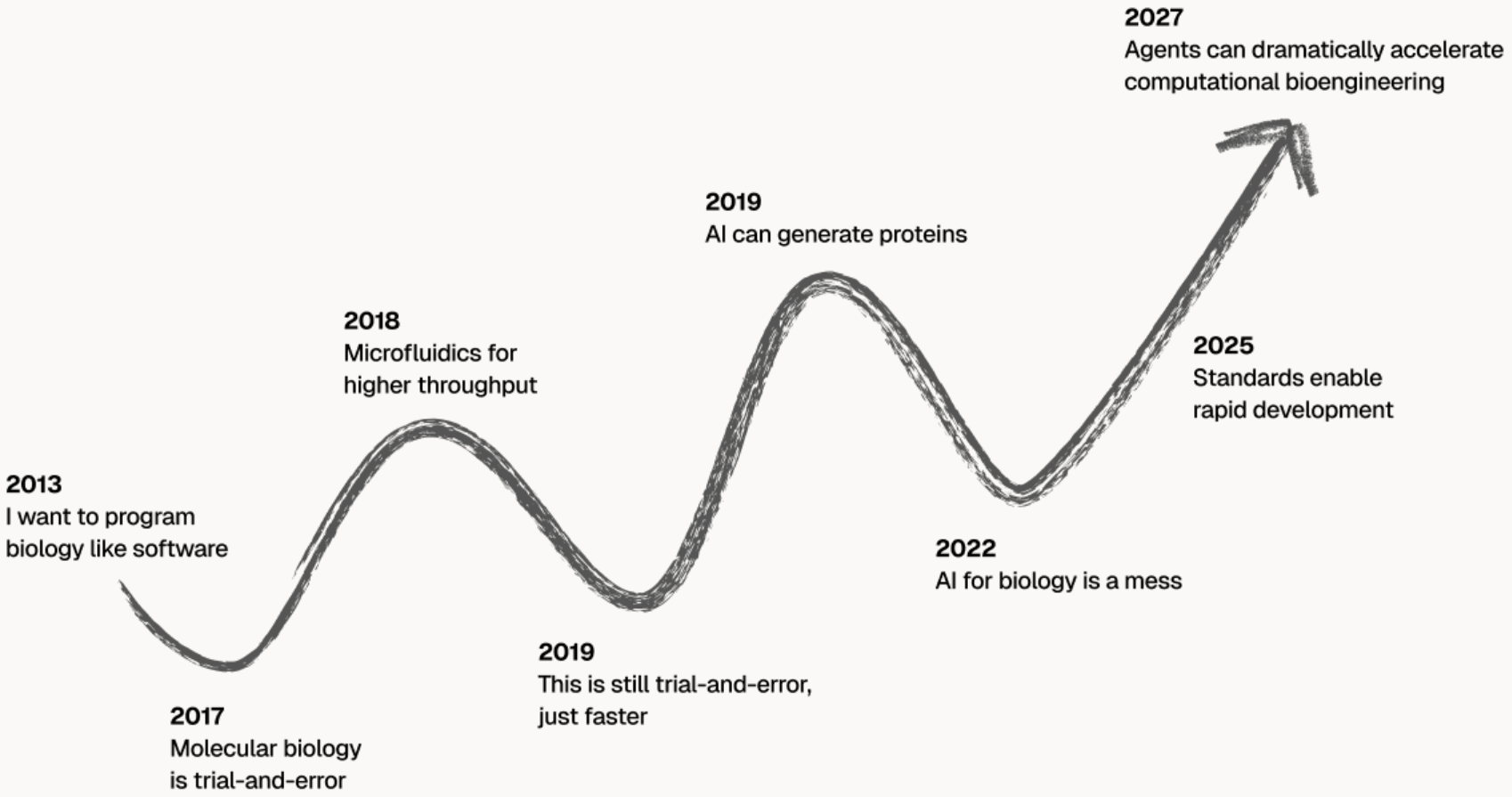
Founding is easy and does not always need much money or time

Be open to opportunities and they will come

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Where are we now with programmable biology?

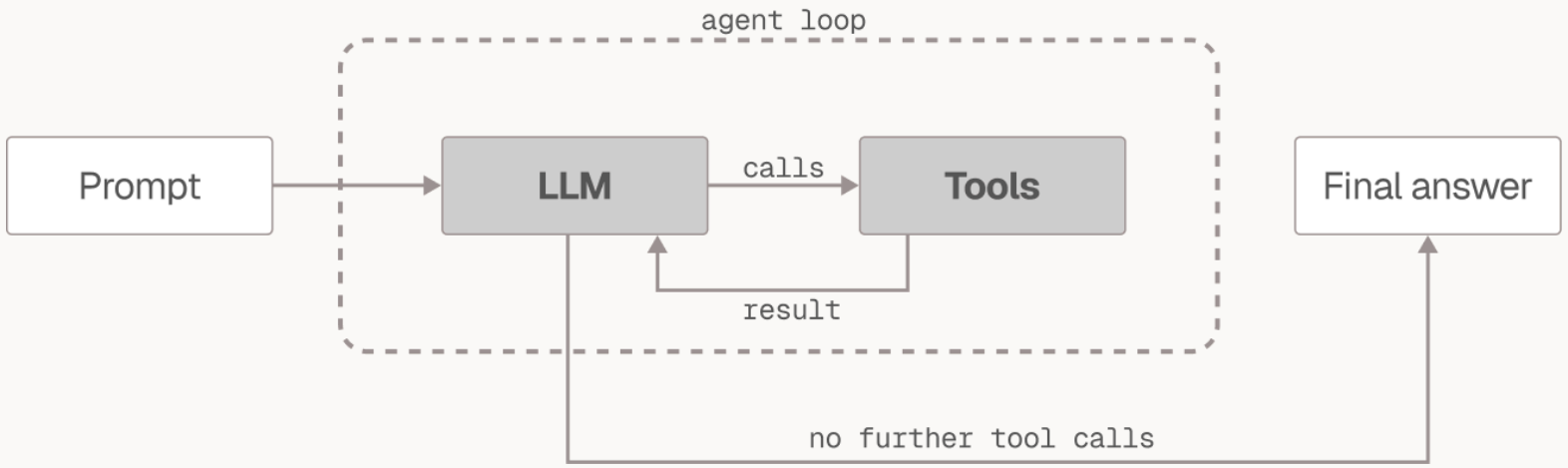
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The agent loop

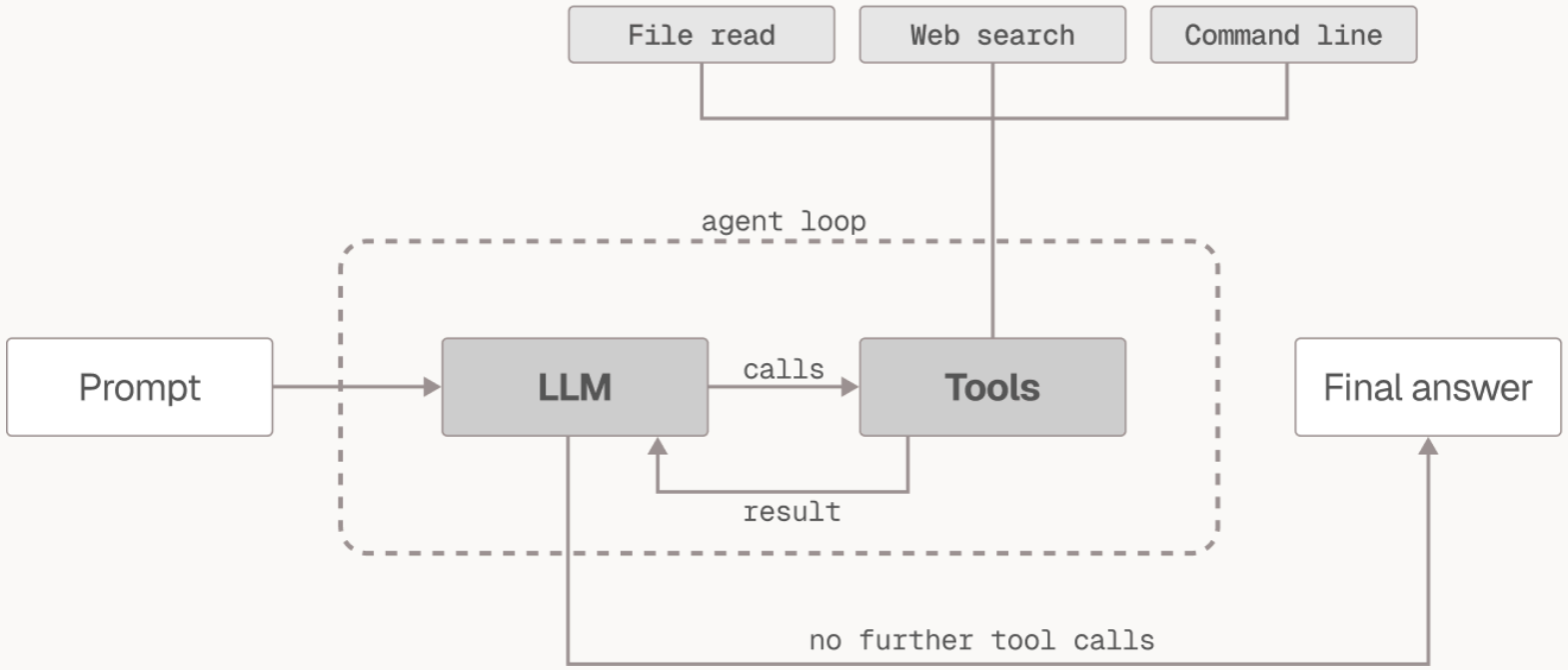
An LLM with tool calls, iterating until a goal is met



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The agent loop

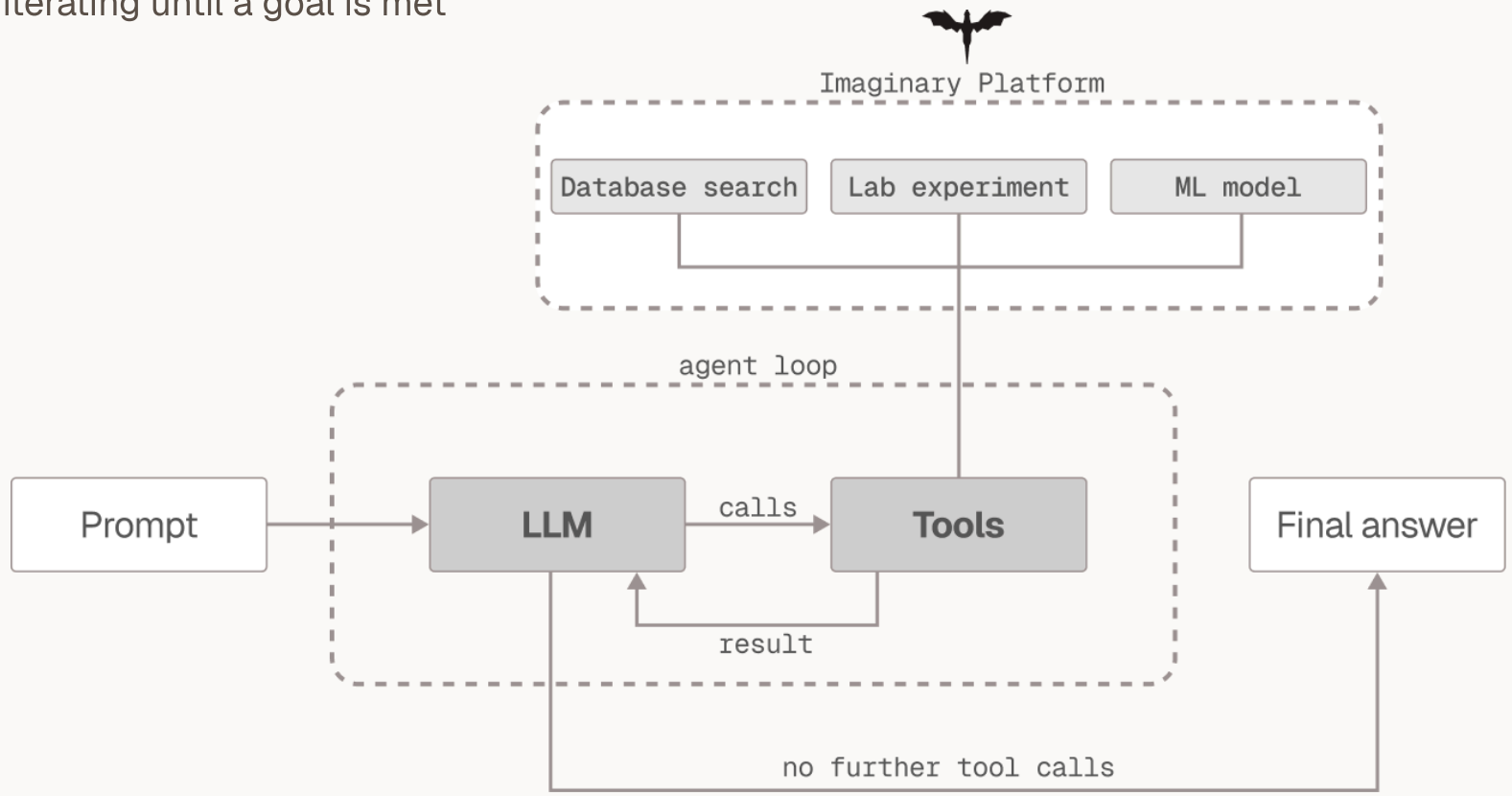
An LLM with tool calls, iterating until a goal is met



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The agent loop for bioengineering

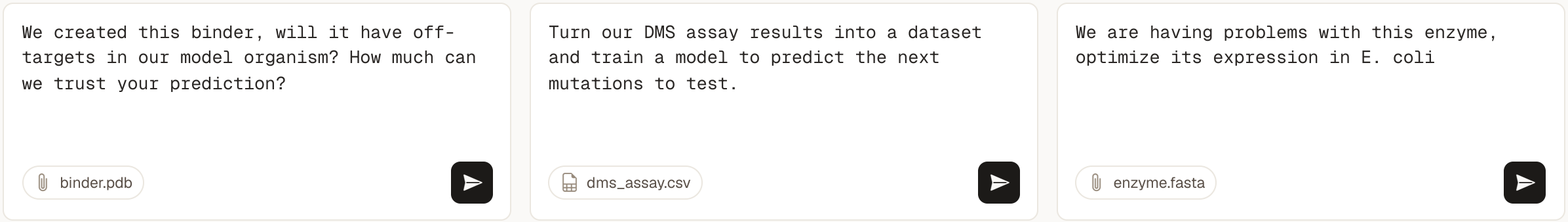
An LLM with tool calls, iterating until a goal is met



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What becomes possible

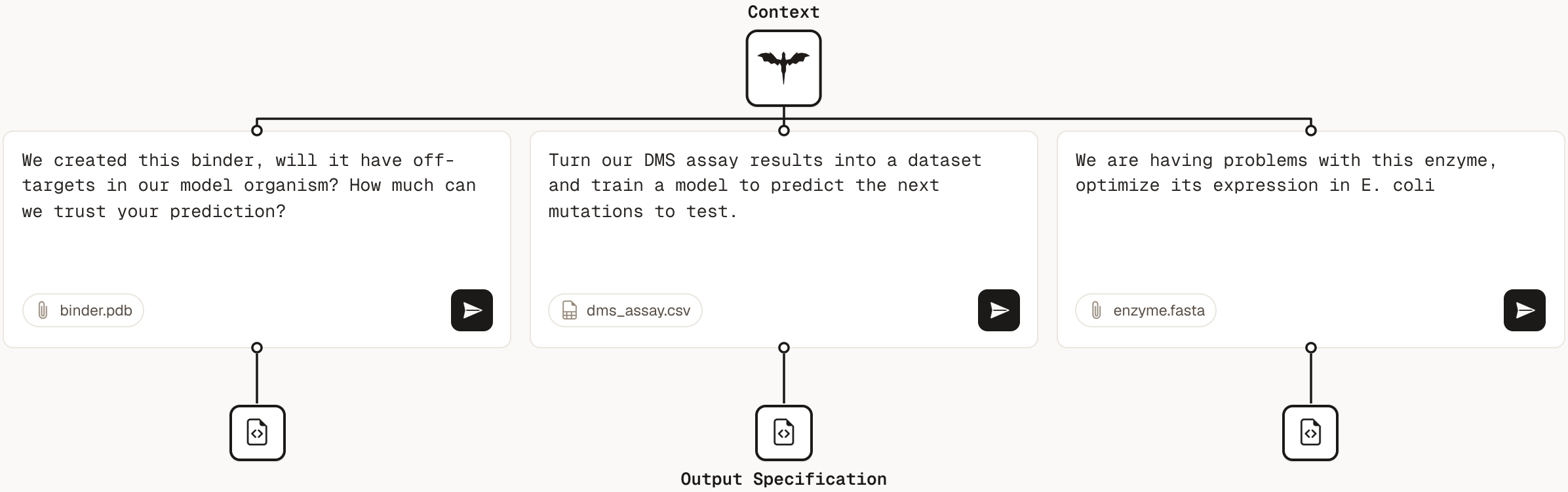
Automated computational bioengineering through natural language



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What becomes possible

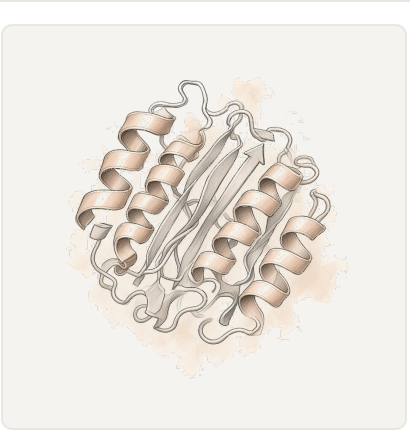
Automated computational bioengineering through natural language



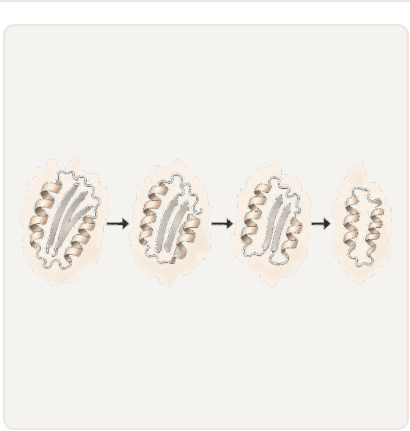
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The long-term vision for Imaginary

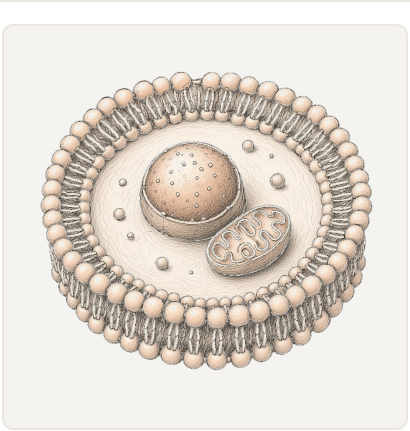
From the molecular level to computational design of biological systems



Molecules



Systems of molecules



Synthetic cells



Multicellular synthetic organisms

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BETA LAUNCH IN JANUARY

...if you would like to test the platform

SYSTEMS BIOLOGY

...if you would like feature support beyond molecules

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...if you would like to join us in making biology programmable

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